⚽ FIFA World Cup Analytics Dashboard

Sector: Sports Analytics

Submitted By

Team ID: Section D – Group 11

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Project Resources

- **GitHub Repository:**
  https://github.com/Silence91169/SectionD_Group11_FiFa
- **Tableau Dashboard:**
  https://public.tableau.com/app/profile/shivam.mishra3999/viz/dva-tableau/Dashboard3

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Executive Summary

This project analyzes historical FIFA World Cup player-level data to uncover patterns in team composition, player roles, and match-level events that influence overall team performance and tournament outcomes. The objective is to support data-driven decision-making for sports analysts, team strategists, and performance evaluators.

The dataset was sourced from Kaggle and includes player participation data across multiple World Cup tournaments. A complete data pipeline was implemented using Python, including data extraction, cleaning, transformation, and analysis using libraries such as pandas, NumPy, and statistical methods. The cleaned dataset was then used to perform exploratory and statistical analysis, followed by visualization design in Tableau to enable interactive insights.

Key findings indicate that player positioning, match participation frequency, and event contributions (such as goals and disciplinary actions) play a significant role in team outcomes. Additionally, patterns in player roles (starters vs substitutes) and team-level consistency highlight strategic differences across teams and tournaments.

Based on the analysis, it is recommended that teams focus on optimizing player positioning strategies, improving consistency in starting lineups, and leveraging data-driven insights to enhance match preparation. The developed dashboard provides an interactive platform for exploring these patterns and supporting informed strategic decisions.

Sector & Business Context

The sports analytics sector has evolved significantly over the past decade, with data playing a critical role in improving team performance, player evaluation, and strategic decision-making. In modern competitive sports such as football, teams increasingly rely on data-driven insights to gain a competitive edge, optimize player utilization, and enhance match outcomes.

In the context of international tournaments like the FIFA World Cup, understanding patterns in player participation, team composition, and match events becomes essential for coaches, analysts, and sports strategists. The decision-makers in this scenario include team management, coaching staff, and performance analysts who are responsible for selecting line-ups, defining strategies, and improving team efficiency.

This project was chosen to explore how historical player-level data can be leveraged to uncover meaningful patterns that influence team success. Despite the availability of large volumes of sports data, it is often underutilized for structured analysis and decision support.

The business value of this project lies in its ability to transform raw historical data into actionable insights. By identifying trends in player roles, participation frequency, and match events, the analysis supports more informed strategic planning, better player selection decisions, and improved understanding of performance dynamics in high-stakes tournaments.

Problem Statement & Objectives

Despite the availability of extensive historical data from FIFA World Cup tournaments, there is a lack of structured analysis that translates player-level data into actionable insights for team strategy and performance optimization. Teams and analysts often rely on fragmented observations rather than comprehensive data-driven evaluation of player roles, participation patterns, and match events.

This project aims to address this gap by analyzing player-level World Cup data to identify key factors that influence team performance, player utilization, and match outcomes.

Objectives

- To analyze player participation patterns across matches and tournaments
- To evaluate the impact of player roles (starter vs substitute) on match dynamics
- To identify trends in player positions and their contribution to match events
- To uncover patterns in key events such as goals and disciplinary actions
- To support strategic insights for team composition and match preparation

Project Scope

In Scope:
- Analysis of player-level data from FIFA World Cup tournaments
- Evaluation of player roles, positions, and match events
- Identification of patterns and trends across matches and teams
- Development of insights to support decision-making

Out of Scope:
- Real-time match prediction or forecasting
- Detailed performance metrics such as assists, passes, or player fitness
- External factors such as weather, injuries, or team finances

Success Criteria

The project will be considered successful if:

- Clear patterns and relationships are identified within the dataset
- Key insights are derived that support strategic decision-making
- KPIs are defined and effectively used in analysis
- The Tableau dashboard enables intuitive and interactive exploration of data
- The analysis is reproducible and well-documented through Python notebooks

Data Description

The dataset used in this project was sourced from Kaggle and contains historical FIFA World Cup player-level data. It includes detailed information about player participation, team composition, and match-level events across multiple tournaments.

Dataset Source:

https://www.kaggle.com/datasets/abecklas/fifa-world-cup

Data Structure

Primary Dataset:WorldCupPlayers.csv
Number of Rows:
Number of Columns:
Granularity:One row per player per match

Key Fields

Some of the key columns used in analysis include:

MatchID: Unique identifier for each match
RoundID: Identifier for tournament stage
Team Initials:Represents the country/team
Player Name:Name of the player
Line-up:Indicates whether the player is a starter or substitute
Position:Player's role on the field (GK, DF, MF, FW)
Event: Records key match events such as goals and disciplinary actions

A detailed column-level explanation is available in the data dictionary located in
https://github.com/Silence91169/SectionD_Group11_FiFa/blob/main/docs/data_dictionary.md

Data Limitations

- The dataset lacks advanced performance metrics such as assists, passes, or player efficiency

- Event data is sparse and only captures major events (goals, cards)
- Missing values are present in columns such as Position and Shirt Number
- The dataset is historical and does not reflect real-time gameplay dynamics
- External factors such as team strategy, injuries, or environmental conditions are not included

Despite these limitations, the dataset provides sufficient information to analyze player participation patterns, team composition, and event-driven insights.

Data Cleaning & ETL Pipeline

All data extraction, cleaning, transformation, and preparation steps were executed using Python in a structured ETL pipeline, as documented across the project notebooks.

Data Extraction

The raw dataset (`WorldCupPlayers.csv`) was loaded using pandas and inspected for structure, completeness, and quality. Initial exploration revealed:

- 37,784 rows and 9 columns
- 736 exact duplicate records
- Significant missing values in Position (89%) and Event (76%) columns
- 82 unique teams, 836 matches, and 7,663 unique players

Data Cleaning

The dataset was cleaned through a series of systematic transformations:

Column Standardization

Column names were converted to snake_case format for consistency and ease of use.

Duplicate Removal

736 duplicate rows (~1.95%) were removed to ensure data integrity.

Whitespace & Formatting

All string columns were stripped of leading/trailing whitespace and standardized.

Handling Missing Values

Position values were standardized into categories such as Outfield and Goalkeeper
Missing Event values were interpreted as no recorded match event
Shirt Number values of 0 (historical placeholder) were converted to null

Categorical Mapping

Line-up values were mapped from short codes (S, N) to descriptive labels (Starting, Substitute)
Position values were normalized and enriched with additional flags

Feature Engineering

A major component of the pipeline involved transforming the raw Event column into structured analytical features:

Extracted event-based columns

` goals`, `penalties_scored`, `own_goals`
`yellow_cards`, `red_cards`, `second_yellow_reds`
`missed_penalties`, `sub_in`, `sub_out`

Created derived metrics

`total_dismissals` = red_cards + second_yellow_reds
Adjusted yellow card counts to avoid double-counting second-yellow dismissals

Generated boolean flags

`is_goalkeeper`, `is_captain`, substitution indicators

Extracted coach nationality from the Coach Name field using regex parsing

Data Transformation

Ensured consistent structure: one row per player per match
Converted all derived fields into appropriate numeric formats
https://github.com/Silence91169/SectionD_Group11_FiFa/blob/main/data/processed/cleaned_worldcup_players.csv

Final dataset:

Rows: 37,048
Columns:22

Assumptions

Missing event values indicate no significant match event occurred
Captaincy and goalkeeper roles are derived from encoded fields
Historical variations in country codes are preserved to maintain dataset authenticity

Reference

Detailed cleaning logic and transformations are documented in:
https://github.com/Silence91169/SectionD_Group11_FiFa/blob/main/notebooks/02_cleaning.ipynb

Kpi & Metric Framework

To evaluate team performance and player contributions, a set of key performance indicators (KPIs) were defined and computed using Python.

KPI Definitions

Team Total Goals
Definition: Total goals scored by a team
Business Relevance: Measures offensive performance
Team Total Yellow Cards
Definition: Total yellow cards received
Business Relevance: Indicates disciplinary behavior
Team Total Red Cards
Definition: Total red cards received
Business Relevance: Reflects critical infractions
Team Total Dismissals
Definition: Sum of red cards and second yellow red cards
Business Relevance: Captures total player removals
Team Matches Played
Definition: Total number of matches played by a team
Business Relevance: Provides context for normalization
Team Unique Players
Definition: Count of distinct players in a team
Business Relevance: Indicates squad rotation and utilization
Goals per Match
Definition: Total goals divided by matches played
Business Relevance: Measures scoring efficiency
Cards per Match
Definition: Total yellow and red cards divided by matches played
Business Relevance: Measures discipline per game

Derived Flags for Analysis

Additional boolean indicators were created to support segmentation and filtering:

- Is Goalscorer
- Is Carded
- Is Starter
- Is Substitute Used
- Is Penalty Taker
- Foreign Coach Indicator

KPI Justification

These KPIs were selected to provide a balanced view of:

- Offensive performance (goals, scoring rate)
- Defensive discipline (cards, dismissals)
- Team composition and participation
- Strategic factors such as substitutions and coaching influence

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to identify patterns, trends, and relationships within the dataset. The analysis focused on player participation, team dominance, scoring behavior, and disciplinary trends across FIFA World Cup tournaments.

Team Participation and Dominance

Analysis of player appearances across countries revealed that teams such as Brazil (BRA), Germany (GER), Argentina (ARG), and Italy (ITA) have consistently high participation levels across tournaments. This indicates sustained performance and deeper squad utilization over time.

Goal Scoring Patterns

The distribution of goals per player-match shows that most players score at most one goal in a match, with very few instances of multiple goals. This suggests that scoring is concentrated among key attacking players rather than evenly distributed across the team.

Player Role Distribution

The dataset shows an approximately balanced distribution between starting players and substitutes. However, starters contribute significantly more to key events such as goals and disciplinary actions, highlighting their greater impact due to higher playtime.

Disciplinary Trends

Teams with higher participation levels tend to accumulate more yellow and red cards. While this is partly due to increased match exposure, it also reflects that high-intensity teams may engage in more aggressive play.

Position Analysis

Outfield players dominate the dataset compared to goalkeepers, reflecting the natural composition of football teams. Goalkeepers represent a smaller but critical subset with specialized roles.

Coaching Influence

Analysis of coaching data indicates that certain countries consistently produce coaches who manage multiple World Cup matches. This suggests the presence of strong football management ecosystems in these regions.

Event-Based Insights

Event analysis shows that most player records do not contain event data, indicating that key match events such as goals and cards are concentrated among a smaller subset of players. Substitutions are also relatively infrequent compared to total player entries.

Player Participation Trends

A small group of players appear in a large number of matches, indicating high experience and consistent selection. These players are likely key contributors to long-term team success.

Correlation Insights

A positive relationship was observed between goals scored and yellow cards at the team level. This suggests that teams with higher offensive activity may also display more aggressive gameplay, leading to increased disciplinary actions.

Summary

Overall, the EDA highlights that player roles, participation frequency, and event contributions are critical factors influencing team performance. These findings provide a strong foundation for deeper statistical analysis.

https://github.com/Silence91169/SectionD_Group11_FiFa/blob/main/notebooks/03_eda.ipynb

Statistical Analysis

Statistical analysis was performed to validate patterns observed during exploratory analysis and to establish data-driven relationships between key variables. Multiple statistical tests and methods were applied to evaluate player behavior, team performance, and strategic factors.

Line-up and Disciplinary Behavior

A Chi-Square test was conducted to evaluate the association between player roles (starting vs substitute) and the likelihood of receiving yellow cards. The results indicate a significant association, with starting players more likely to receive yellow cards. This can be attributed to longer playtime and greater involvement in match situations.

Goal Scoring Comparison (Starters vs Substitutes)

A Mann-Whitney U test was performed to compare goal-scoring behavior between starting players and substitutes. The analysis shows that starters score significantly more goals on average compared to substitutes. This reinforces the importance of starting lineup selection in determining match outcomes.

Team-Level Correlation Analysis

Correlation analysis using Pearson and Spearman methods was conducted across team-level metrics such as total goals, yellow cards, red cards, and matches played. A strong positive correlation was observed between total goals and yellow cards. However, this relationship is influenced by the number of matches played, indicating that more active teams tend to accumulate both higher goals and disciplinary actions.

Captaincy and Discipline

A Mann-Whitney U test was used to examine whether captains exhibit different disciplinary behavior compared to non-captain players. The results suggest that captains tend to receive more yellow cards, possibly due to their increased involvement in gameplay and decision-making situations.

Coaching Influence on Performance

The impact of foreign versus domestic coaches on team performance was analyzed using statistical comparison tests. The results indicate variation in goal-scoring patterns depending on coaching nationality, suggesting that coaching strategy and background may influence team performance.

Goals per Match Efficiency

Descriptive analysis was used to rank teams based on goals per match, considering teams with a minimum threshold of matches played. This provided insights into the most efficient scoring teams across tournaments.

Summary

The statistical analysis validates key observations from the EDA and provides deeper insights into player roles, team behavior, and performance dynamics. These findings support the development of data-driven strategies and reinforce the importance of structured analysis in understanding complex sports datasets.

https://github.com/Silence91169/SectionD_Group11_FiFa/blob/main/notebooks/04_statistical_analysis.ipynb

Tableau Dashboard Design

An interactive Tableau dashboard was developed to visualize key insights derived from the FIFA World Cup dataset. The dashboard is designed using a dark-themed interface with clear visual hierarchy, enabling intuitive exploration of player, team, and match-level patterns.

Dashboard Structure

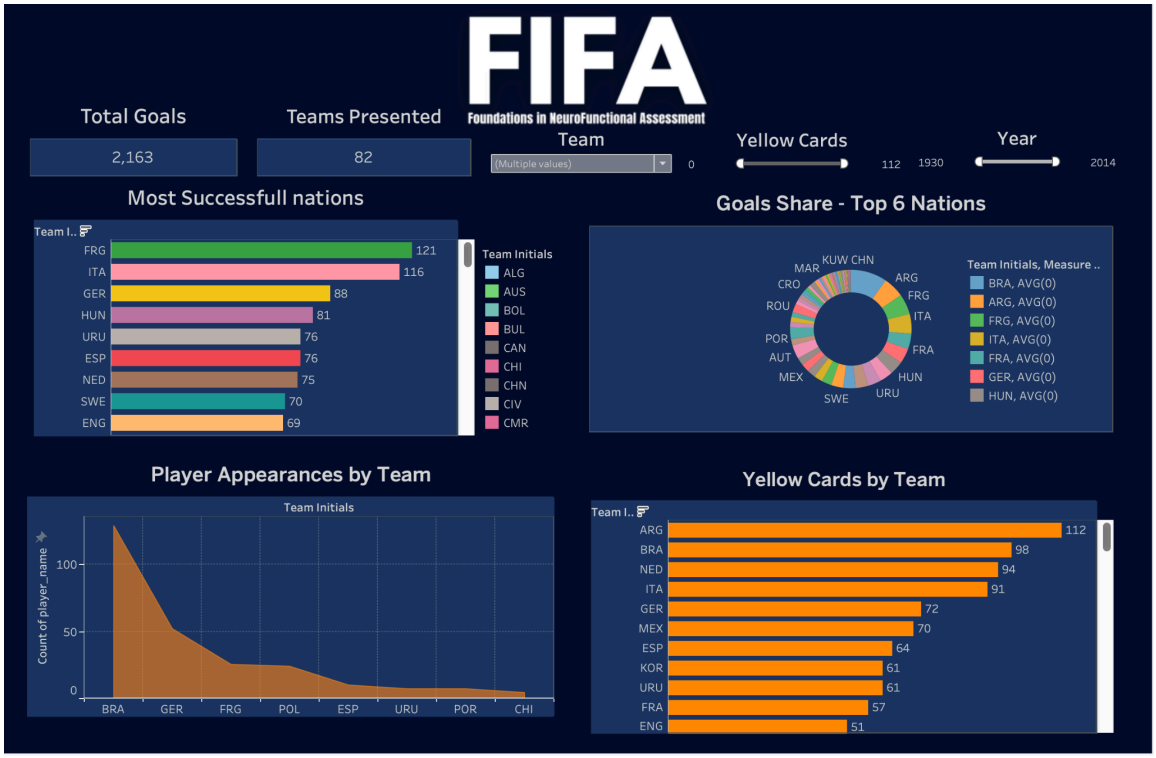
The dashboard is organized into three main sections, each focusing on a specific analytical perspective:

1. **Tournament & Team Overview**
2. **Player-Level Analysis**
3. **Substitution & Discipline Analysis**

This multi-screen structure ensures logical flow and allows users to explore insights at different levels of granularity.

Dashboard 1: Tournament & Team Overview

This dashboard provides a high-level overview of tournament performance and team dominance.



Key Components:

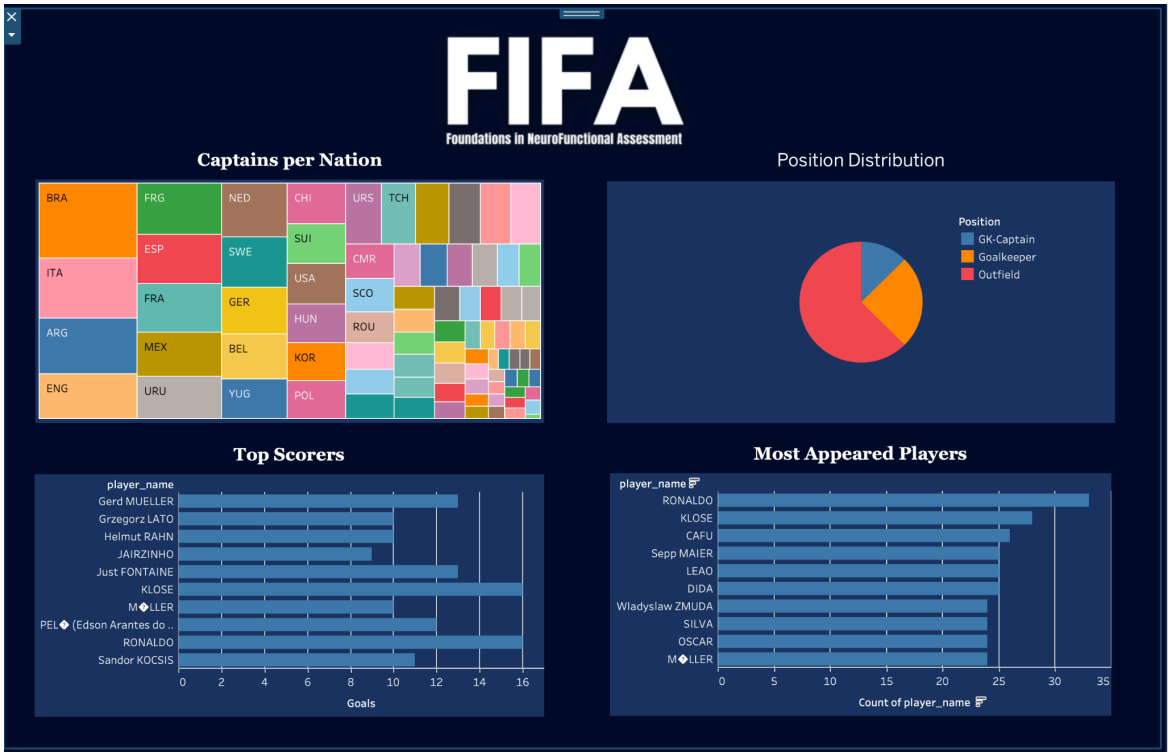
- **KPI Cards:** Total goals and number of teams presented
- **Bar Chart (Most Successful Nations):** Highlights historically dominant teams such as Brazil and Germany
- **Donut Chart (Goals Share):** Shows contribution of top-performing nations
- **Area Chart (Player Appearances):** Represents participation trends across teams
- **Bar Chart (Yellow Cards):** Displays disciplinary behavior by team

Insight Focus:

This dashboard helps identify dominant teams, scoring distribution, and overall tournament trends.

Dashboard 2: Player-Level Analysis

This dashboard focuses on player-level insights and team composition.



Key Components:

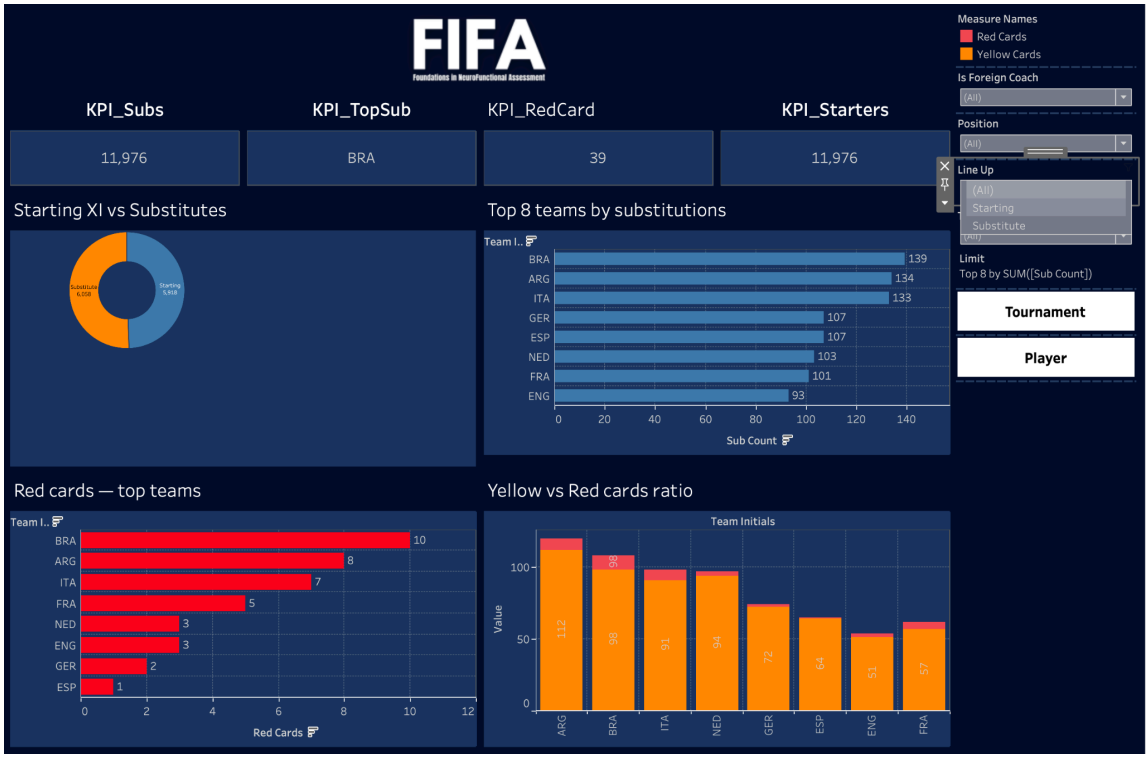
- **Treemap (Captains per Nation):** Shows leadership distribution across countries
- **Pie Chart (Position Distribution):** Highlights proportion of goalkeepers vs outfield players
- **Bar Chart (Top Scorers):** Displays players with highest goal contributions
- **Bar Chart (Most Appeared Players):** Identifies players with highest match participation

Insight Focus:

This dashboard enables analysis of player roles, experience, and contribution to team performance.

Dashboard 3: Substitution & Discipline Analysis

This dashboard explores substitution patterns and disciplinary trends.



Key Components:

- **KPI Cards:** Substitutions, top substituting team, red cards, starters count
- **Donut Chart (Starting XI vs Substitutes):** Compares player roles
- **Bar Chart (Top Teams by Substitutions):** Identifies teams with highest substitution activity
- **Bar Chart (Red Cards by Team):** Highlights disciplinary extremes
- **Stacked Bar Chart (Yellow vs Red Cards):** Compares card distribution across teams

Insight Focus:

This dashboard highlights team strategies related to substitutions and disciplinary behavior.

Filters and Interactivity

The dashboard includes multiple interactive filters to enhance user exploration:

- Team filter
- Year filter (range slider from 1930 to 2014)
- Position filter
- Line-up filter (Starter/Substitute)

- Foreign coach indicator

These filters allow users to dynamically adjust the view and analyze specific segments of the data.

Design Choices

- A **dark theme** was used to improve visual contrast and readability
- Bright accent colors were applied to highlight key metrics and categories
- Consistent layout and alignment were maintained across all screens
- Visual elements were selected based on the nature of data (distribution, comparison, ranking)

Dashboard Objective

The dashboard transforms complex player-level data into an interactive visual storytelling platform. It enables users to quickly identify patterns in team dominance, player contributions, substitutions, and disciplinary trends.

https://public.tableau.com/views/dva-tableau/Dashboard3?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Insights Summary

1. Teams with consistently high player participation, such as Brazil, Germany, Argentina, and Italy, demonstrate sustained dominance across tournaments, indicating strong squad depth and long-term performance stability.
2. Goal scoring is highly concentrated among a limited number of players, suggesting that teams rely heavily on key attacking individuals rather than distributed scoring strategies.
3. Starting players contribute significantly more to both offensive and defensive match events compared to substitutes, highlighting the critical importance of optimal starting lineup selection.
4. Teams with higher match participation tend to accumulate more yellow and red cards, indicating that aggressive gameplay and extended tournament exposure increase disciplinary risks.
5. A strong positive relationship exists between total goals and yellow cards at the team level, suggesting that offensively active teams often engage in high-intensity play that leads to more fouls.
6. Captains exhibit higher disciplinary involvement compared to non-captain players, likely due to their leadership role and increased engagement in match situations.
7. Substitution events are relatively infrequent compared to total player appearances, indicating that most match participation is driven by starting players rather than dynamic substitutions.

8. A small subset of players consistently appears across a large number of matches, indicating high experience levels and long-term strategic importance within teams.
9. Coaching patterns reveal that certain countries consistently produce coaches with high match involvement, suggesting structured development systems for football management.
10. Variations in goal-scoring patterns based on coaching nationality indicate that coaching background and strategy may influence team performance outcomes.

Recommendations

1. Optimize Starting Lineup Selection

Teams should prioritize data-driven selection of starting players, as starters have significantly higher impact on goals and match events. Ensuring the strongest possible starting lineup can improve overall match outcomes.

2. Focus on Key Goal-Scoring Players

Since goal contributions are concentrated among a few players, teams should identify and support high-performing attackers. Strategic positioning and resource allocation around these players can enhance scoring efficiency.

3. Improve Disciplinary Control Strategies

Teams should implement strategies to reduce unnecessary fouls and card accumulation, especially for high-intensity teams. Lower disciplinary rates can prevent player dismissals and maintain competitive advantage.

4. Leverage Experienced Players Strategically

Players with high match participation should be utilized effectively in critical matches due to their experience and consistency. Their presence can positively influence team performance and decision-making under pressure.

5. Evaluate Coaching Strategies and Diversity

Teams should analyze the impact of coaching backgrounds and consider diverse coaching strategies where beneficial. Exposure to different tactical approaches may improve team adaptability and performance.

6. Enhance Substitution Strategies

Given the limited impact of substitutions observed in the data, teams should reassess substitution timing and effectiveness to maximize their influence during matches.

Impact Estimation

The insights and recommendations derived from this analysis can lead to measurable improvements in team performance, strategic decision-making, and overall efficiency in competitive environments.

Performance Improvement

By optimizing starting lineups and focusing on high-impact players, teams can potentially improve goal-scoring efficiency. Even a modest increase of 5–10% in scoring efficiency could significantly influence match outcomes in tournament settings.

Reduction in Disciplinary Risks

Implementing better disciplinary control strategies can reduce yellow and red card accumulation. A reduction in dismissals can help teams maintain full squad strength during critical matches, thereby improving win probability and match stability.

Strategic Player Utilization

Leveraging experienced players more effectively can enhance decision-making during high-pressure situations. This can lead to improved match control and better performance in knockout stages, where experience plays a crucial role.

Coaching Strategy Optimization

Insights related to coaching patterns can support teams in evaluating and refining their coaching strategies. Adoption of effective tactical approaches may improve team adaptability and overall performance.

Improved Match Preparation

The ability to analyze historical patterns in player roles and match events enables teams to prepare more effectively for different match scenarios. This can result in better in-game decisions and tactical adjustments.

Overall Value

While exact numerical impact may vary depending on team context and implementation, the analysis demonstrates how structured data-driven approaches can provide a competitive

advantage. The application of these insights can lead to improved performance consistency, reduced risks, and more informed strategic planning.

Future Scope

While this project provides meaningful insights into player participation and team behavior in FIFA World Cup tournaments, several opportunities exist to extend and enhance the analysis further.

Inclusion of Advanced Performance Metrics

Future work can incorporate detailed player-level statistics such as assists, passes, distance covered, and defensive actions. This would enable a more comprehensive evaluation of player performance and team strategies.

Predictive Modeling

Machine learning models can be developed to predict match outcomes, player performance, or goal-scoring probabilities based on historical data. This would transform the analysis from descriptive to predictive analytics.

Real-Time Data Integration

Integrating live match data can enable real-time dashboards and dynamic decision support systems. This would allow analysts and teams to make data-driven decisions during ongoing matches.

Enhanced Player Segmentation

Clustering techniques can be applied to group players based on performance, roles, and behavioral patterns. This would help in identifying player archetypes and optimizing team composition.

Expanded Dataset Integration

Combining this dataset with additional sources such as match-level data, team statistics, and external factors (weather, injuries, etc.) would provide a more holistic view of match dynamics.

Advanced Visualization and Dashboarding

Future iterations can include more interactive and multi-layered dashboards with advanced filtering, drill-down capabilities, and scenario-based analysis to support deeper exploration.

Tactical and Strategic Analysis

Further analysis can focus on tactical aspects such as formation patterns, substitution timing, and in-game decision-making strategies to provide more actionable insights for coaching staff.

Conclusion

This project demonstrates how historical FIFA World Cup data can be transformed into meaningful insights through structured data analysis and visualization. By implementing a complete data pipeline and leveraging statistical methods, the analysis identified key patterns in player participation, team performance, and match dynamics.

The findings highlight the importance of starting lineup selection, goal-scoring concentration among key players, and the impact of disciplinary behavior on team performance. The Tableau dashboard further enhances these insights by providing an interactive platform for exploration and decision-making.

Overall, the project illustrates the value of data-driven approaches in sports analytics and showcases how analytical techniques can be applied to derive actionable insights in competitive environments.

Appendix

- **GitHub Repository:**
  https://github.com/Silence91169/SectionD_Group11_FiFa
- **Data Dictionary:**
  Available in `/docs/data_dictionary.md`
- **Notebooks:**
  - 01_extraction.ipynb
  - 02_cleaning.ipynb
  - 03_eda.ipynb
  - 04_statistical_analysis.ipynb
  - 05_final_load_prep.ipynb
- **Processed Dataset:**
  `data/processed/cleaned_worldcup_players.csv`
- **ETL Pipeline Script:**
  `scripts/etl_pipeline.py`
- **Tableau Dashboard:**
  https://public.tableau.com/views/dva-tableau/Dashboard3?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

Contribution Matrix

Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau Dashboard	Report Writing	PPT
Palak						☑	☑
Shitanshu	☑	☑	☑	☑	☑		☑
Krishna					☑		
Shivam					☑		
Aditya		☑	☑	☑	☑		
Prajjwal		☑					